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Form No – 260605553

Q1 :-

Code

```
#include <iostream>
using namespace std;

class Student {
    int rollNumber;
    string name;
    double percentage;
    static int totalStudents;

    Student() {
        rollNumber = 0;
        name = "";
        percentage = 0;
        totalStudents=0;
    }

    Student(int roll, const string& studentName) {
        rollNumber = roll;
        name = studentName;
    }

    Student (const Student &obj1) {
        cout<<"Copy constructor called";
    }

    void setPercentage(double percent) {
        if(percent>=0 && percent<=100) {
            percent = percentage;
        }
    }

    double getPercentage() const{
        return percentage;
    }

    char calculateGrade() const {
        if(grade>=80) {
            return "A";
        }
        else if(grade>=70) {
            return "B";
        }
    }
}
```

```

        else if(grade>=60) {
            return "C";
        }

        else if(grade>=50) {
            return "D";
        }

        else {
            return "F";
        }

    }

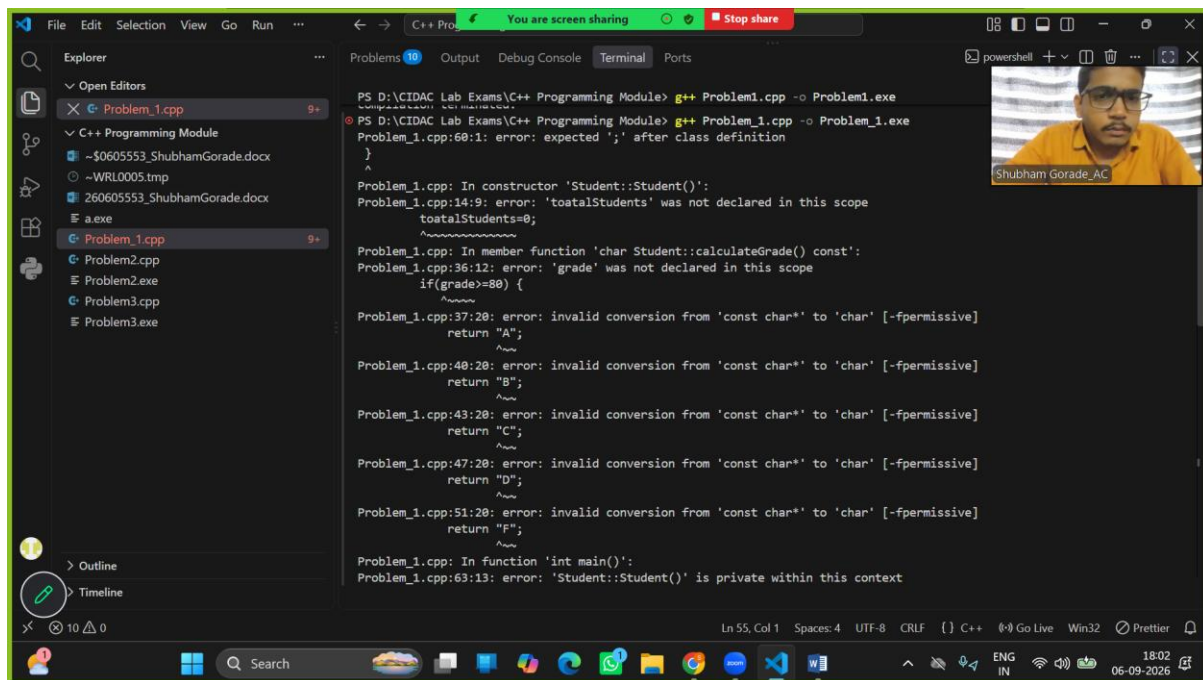
    ~Student() {
        cout<<"Destructor called";
    }
}

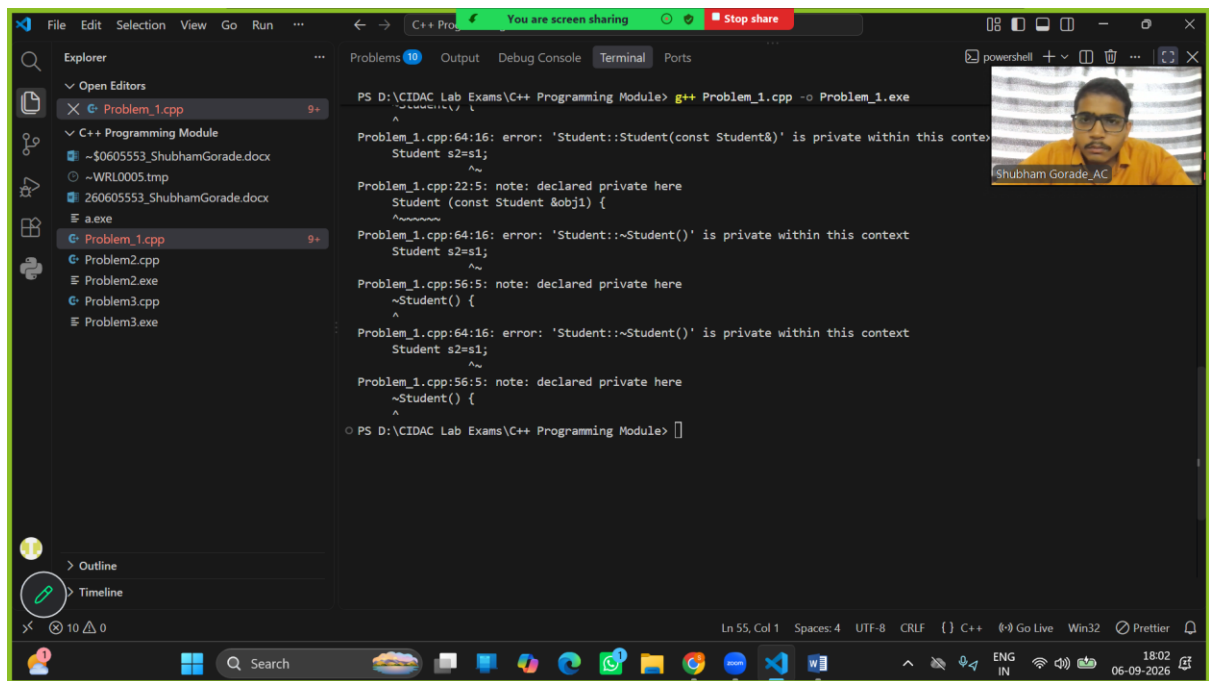
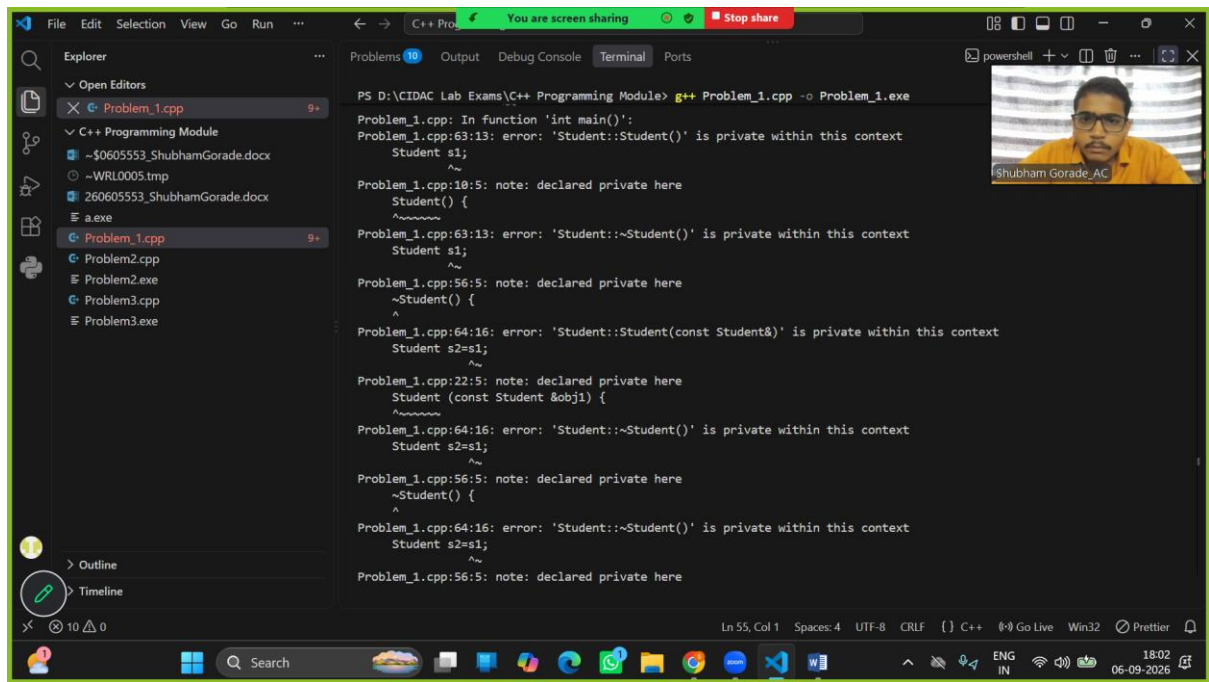
int main() {
    Student s1;
    Student s2=s1;

    return 0;
}

```

Output





Q2:-

Code

```
#include <iostream>
using namespace std;

class Employee {
    int empId;
    string name;
    double salary;
    int performanceRating;

public:
    Employee() {
        empId=0;
        name="";
        salary=0;
        performanceRating=0;
    }

    Employee (int id, const string& empName, double sal, int rating){
    }

    void accept() {
        cin>>empId>>name>>salary>>performanceRating;
    }

    double calculateBonus() const {
        if(performanceRating==1) {
            return (salary*0)/100;
        }
        else if(performanceRating==2) {
            return (salary*5)/100;
        }
        else if(performanceRating==3) {
            return (salary*10)/100;
        }
        else if(performanceRating==4) {
            return (salary*15)/100;
        }
        if(performanceRating==5) {
            return (salary*20)/100;
        }
        else{
            cout<<"Please enter valid reading in between 1 to 5";
            return 0;
        }
    }
}
```

```

        void displayDetails() const{
            cout<<"ID: "<<empId<<" | Name:"<<name<<" | Salary: "<<salary<<" |
Rating:"<<performanceRating<<" | Bonus: "
            <<calculateBonus()<<endl;
        }
};

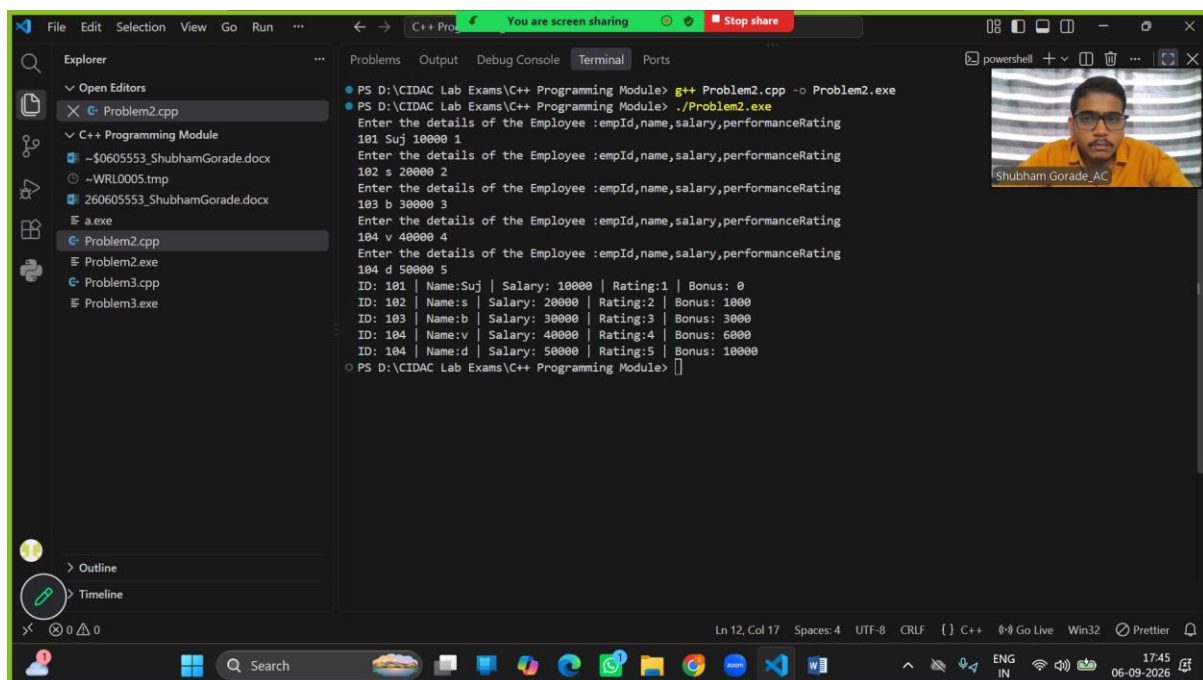
int main() {
    Employee obj[5];

    for(int i=0; i<5; i++){
        cout<<"Enter the details of the Employee
:empId,name,salary,performanceRating"<<endl;
        obj[i].accept();
    }

    for(int i=0; i<5; i++){
        obj[i].displayDetails();
    }
    return 0;
}

```

Output



The screenshot shows a C++ IDE with the following components:

- Explorer:** Lists files including `Problem2.cpp`, `Problem2.exe`, `Problem3.cpp`, and `Problem3.exe`.
- Terminal:** Shows the execution of `g++ Problem2.cpp -o Problem2.exe` and the program's output. The output prompts for employee details and displays the following table:

ID	Name	Salary	Rating	Bonus
101	Suj	10000	1	0
102	s	20000	2	1000
103	b	30000	3	3000
104	v	40000	4	6000
104	d	50000	5	10000

The terminal also shows the command prompt `PS D:\CIDAC Lab Exams\C++ Programming Module>` at the bottom.

Q3:-

Code

```
#include <iostream>
using namespace std;

class Tempreature {
public :
    double celsius;

    Tempreature() {
        celsius=0;
    }

    Tempreature(double cel) {
        celsius=cel;
    }

    double toFahrenheit() const {
        return (celsius*9/5)+32;
    }

    double toKelvin() const {
        return celsius+273.15;
    }

    void displayAll() const {
        cout<<"Tempreature : "<<celsius<<"C = "<<toFahrenheit()<<"F = "
        <<toKelvin()<<"K"<<endl;
    }
};

int main() {

    int tt1;
    cout<<"Enter tempreature in celsius";
    cin>>tt1;
    if(tt1>=-273.15){
        Tempreature t1(tt1);
        t1.displayAll();
    }
    else {
        cout<<"ERROR: Temperature below absolute zero (-273.15°C)";
    }

    int tt2;
    cout<<"Enter tempreature in celsius";
```

```

cin>>tt2;
if(tt2>=-273.15){
    Tempreature t2(tt2);
    t2.displayAll();
}
else {
    cout<<"ERROR: Temperature below absolute zero (-273.15°C)";
}

int tt3;
cout<<"Enter tempreature in celsius";
cin>>tt3;
if(tt3>=-273.15){
    Tempreature t3(tt3);
    t3.displayAll();
}
else {
    cout<<"ERROR: Temperature below absolute zero (-273.15°C)";
}

int tt4;
cout<<"Enter tempreature in celsius";
cin>>tt4;
if(tt4>=-273.15){
    Tempreature t4(tt4);
    t4.displayAll();
}
else {
    cout<<"ERROR: Temperature below absolute zero (-273.15°C)";
}

return 0;
}

```

Output

